

Nhat-Minh Nguyen, Ph.D.

Cosmologist, Kavli IPMU Fellow

Kavli Institute for Physics and Mathematics of the Universe, University of Tokyo

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RESEARCH PROFILE

Cosmologist specializing in galaxy surveys, growth of large-scale structure, and statistical methods for cosmological inference. Research experience spans Roman/Euclid-relevant likelihood validation, field-level inference, effective-field-theory likelihoods, multi-tracer analyses, beyond-two-point statistics, and machine-learning methods for survey data. Current and former survey affiliations include PFS and DESI. Builds reproducible analysis paths at the interface between theory, likelihoods, and survey data.

SELECTED HONORS

Buchalter Cosmology Prize , with American Astronomical Society award announcement	2024
JSPS KAKENHI Grant-in-Aid for Research Activity Start-up, PI, JPY 2,000,000	2025
Japan Foundation for Promotion of Astronomy Research Support Grant, PI, approx. JPY 320,000	2025

RESEARCH PROJECTS AND EXPERTISE

- **Galaxy clustering at the field level:** first-author PRL study on how much information can be extracted from galaxy clustering at the field level.
- **EFT likelihoods and physical data models:** work on consistency tests of field-level inference with the EFT likelihood and on the impact of physical data models in forward inference.
- **PFS full-shape pipeline and multi-survey calibration:** responsible for the PFS full-shape galaxy-clustering analysis pipeline; single-author 2026 arXiv preprint, submitted to JCAP, “Multi-tracers, multi-surveys: a joint Fisher analysis of DESI+PFS.”
- **Growth of structure and dark-sector tests:** first-author PRL evidence for suppressed structure growth and co-authored work on Horndeski growth-rate parameterization.
- **Beyond-two-point survey statistics:** contribution to a parameter-masked mock data challenge for beyond-two-point galaxy clustering statistics.
- **Machine learning for reconstruction:** project on reconstructing the local density field with combined convolutional and point-cloud architecture.
- **CMB–large-scale-structure connections:** first-author kSZ forward-modeling work including uncertainties from velocity reconstruction.

TECHNICAL AND SURVEY EXPERIENCE

Surveys	Prime Focus Spectrograph (PFS), Dark Energy Spectroscopic Instrument (DESI), SDSS-III BOSS.
Methods	Field-level inference, forward modeling, EFT likelihoods, multi-tracer analysis, density reconstruction, beyond-two-point statistics, likelihood validation, machine-learning reconstruction.
Science	Large-scale structure, galaxy clustering, intrinsic alignment, growth of structure, modified gravity, dark energy, primordial non-Gaussianity, kinematic Sunyaev-Zel’dovich effect.

POSITIONS

Kavli IPMU Fellow <i>Kavli Institute for Physics and Mathematics of the Universe, University of Tokyo</i>	<i>Oct. 2024 – present</i>
Leinweber Fellow <i>Leinweber Center for Theoretical Physics and Physics Department, University of Michigan</i>	<i>Jan. 2022 – Jul. 2024</i>

Research Associate
Physical Cosmology Group, Max Planck Institute for Astrophysics

Jul. 2020 – Dec. 2021

Ph.D. Candidate
Physical Cosmology Group, Max Planck Institute for Astrophysics

Oct. 2016 – Jun. 2020

EDUCATION

Ph.D., Astrophysics, Magna Cum Laude *2016 – 2020*
International Max Planck Research School on Astrophysics; Ludwig Maximilians University of Munich; Max Planck Institute for Astrophysics

M.Sc., Astronomy and Astrophysics *2014 – 2016*
Erasmus Joint Master Degree in Astronomy & Astrophysics, with full-tuition scholarship and stipend

B.Sc., Physics and Theoretical Physics *2008 – 2013*
Ho Chi Minh University of Science, with honors and scholarships

SELECTED PUBLICATIONS

1. Yu, **Nguyen**. “How I stop worrying about non-universality and b_ϕ : Constraining local f_{NL} with b_ϕ priors from HOD posteriors.” [arXiv preprint](#), 2026.
2. **Nguyen**. “Physics Is All You Need? A Case Study in Physicist-Supervised AI Development of Scientific Software.” [Single-author arXiv preprint](#). Accepted for ICML 2026 AI for Science Workshop.
3. **Nguyen**. “Multi-tracers, multi-surveys: a joint Fisher analysis of DESI+PFS.” [Single-author arXiv preprint](#). Submitted to JCAP, 2026.
4. **Nguyen**, Akitsu, Taruya. “Galaxy sizes as complementary (zero-)bias tracers of local primordial non-Gaussianity.” [arXiv preprint](#). Submitted to PRD, 2026.
5. Barthe-Gold, **Nguyen**, Thiele. “Reconstructing the local density field with combined convolutional and point cloud architecture.” [arXiv preprint](#). Accepted for NeurIPS 2025 Machine Learning for Physical Science Workshop.
6. Beyond-2point Collaboration. “A parameter-masked mock data challenge for beyond-2pt galaxy clustering statistics.” [ApJ 990, 99, 2025](#).
7. **Nguyen**, Schmidt, Tucci, Reinecke, Kostić. “How much information can be extracted from galaxy clustering at the field level?” [Phys. Rev. Lett. 133, 221006, 2024](#).
8. Wen, **Nguyen**, Huterer. “Sweeping through Horndeski Canvas: a New Growth-Rate Parameterization for Modified-Gravity Theories.” [JCAP 09, 028, 2023](#).
9. **Nguyen**, Huterer, Wen. “Evidence for suppression of structure growth in the concordance cosmological model.” [Phys. Rev. Lett. 131, 111001, 2023](#). Editor’s Suggestion.
10. Kostić, **Nguyen**, Schmidt, Reinecke. “Consistency tests of field level inference with the EFT likelihood.” [JCAP 07, 063, 2023](#).
11. Tsaprazi, **Nguyen**, et al. “Field-level inference of galaxy intrinsic alignment from the SDSS-III BOSS survey.” [JCAP 08, 003, 2022](#).
12. **Nguyen** et al. “Impacts of the physical data model on the forward inference of initial conditions from biased tracers.” [JCAP 03, 058, 2021](#).
13. **Nguyen** et al. “Taking measurements of the kinematic Sunyaev-Zel’dovich effect forward: including uncertainties from velocity reconstruction with forward modeling.” [JCAP 12, 011, 2020](#).
14. DESI Collaboration. “DESI 2024 VII: Cosmological Constraints from the Full-Shape Modeling of Clustering Measurements.” [JCAP 07, 028, 2025](#).
15. DESI Collaboration. “DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations.” [JCAP 02, 021, 2025](#).
16. Schmidt et al. “A rigorous EFT-based forward model for large-scale structure.” [JCAP 01, 042, 2019](#).

SELECTED TALKS

Conference and workshop talks only; ordinary seminar talks are intentionally omitted.

2025	Particles, Gravitation and the Universe: from Quantum Mechanics to Quantum Gravity, IOP, VAST, Ha Noi, Viet Nam.
2024	“Theory and Data Analysis Challenges for Cosmological Large-Scale Structure Observations,” YITP, Kyoto, Japan.
2024	COSMO’24, Kyoto, Japan.
2024	PASCOS 2024, Plenary Session, ICISE Quy Nhon, Viet Nam.
2024	Cosmology from Home, online conference.
2024	DESI Research Forum.
2024	Aspen workshop, “Fundamental Physics in the Era of Big Data and Machine Learning.”
2024	“Fundamental Physics from Future Spectroscopic Surveys,” LBNL, Berkeley.
2023	Future Science with CMB x LSS workshop, YITP, Kyoto.
2022	Aspen workshop, “Large-Scale Structure Cosmology beyond 2-Point Statistics.”

MENTORING, TEACHING, AND SERVICE

- Co-mentored Carter Matties at the University of Michigan and Andrija Kostić at the Max Planck Institute for Astrophysics; shared field-level/EFT work with Kostić appears in the PRL 2024 and JCAP 2023 publications listed above.
- Advised Baptiste Barthe-Gold’s density-field reconstruction project, accepted to the NeurIPS 2025 Machine Learning for Physical Science Workshop, and Eleni Tsaprazi’s intrinsic-alignment project, published in JCAP 2022; also advised Andrew Hope, Kyle Lee, Disha Saxena, and Kasey Thai.
- Organizing Committee Chair and lecturer, VSOA10: Cosmology, 10th Vietnam School of Astrophysics, ICISE Quy Nhon, 2026.
- Course instructor, Michigan Math and Science Scholars, Summer 2024; guest lecturer, Summer 2023.
- Organizing Committee Chair, Beyond-2-Point Statistics Meet Survey Systematics, 2025.
- Large-scale Structure Session Convener, COSMO’24, 2024.
- Active referee for Physical Review Letters, Astrophysical Journal Letters, Physical Review D, Journal of Cosmology and Astroparticle Physics, and Astronomy & Astrophysics.

POPULAR SCIENCE AND MEDIA

2024	Vietnamese-language article on DESI and dark energy, Tia Sang Magazine.
2023	“Growing in molasses: Cosmic large-scale structure caught growing slower than expected,” Science X Dialog, Phys.org.
Media	Coverage in Scientific American, New Scientist, VICE, Space.com, Quanta Magazine, University of Michigan News, and Max Planck Institute for Astrophysics Research Highlights.

REFERENCES

Dr. Fabian Schmidt Ph.D. advisor Max Planck Institute for Astrophysics fabians@MPA-Garching.MPG.DE	Prof. Dragan Huterer Postdoctoral advisor University of Michigan huterer@umich.edu	Prof. Elisabeth Krause External collaborator and mentor University of Arizona krausee@arizona.edu
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